

Asian HLA pangenome

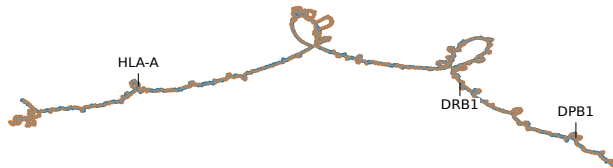
754 haplotype entries, including 462 Asian / Arab entries; 371 distinct donors



HPRC includes the 16 Japanese donors. Five donors overlap between projects; GRCh38 and CHM13 add two reference haplotypes.

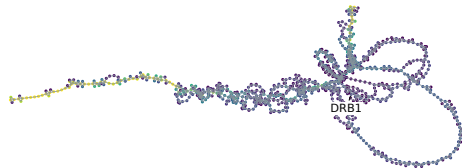
Source	Donor location	Samples
HPRC r2	Worldwide, incl. Japan	232
	Japanese subset (JPT)	16
APR	UAE	53
JaSaPaGe	Saudi Arabia / Japan	9 / 10
K-PanRef	Korea	14
CPC	China	58

MHC construction backbone



1,677 segments; blue: GRCh38, orange: added sequence
Read resource: 1000 Genomes ($n = 2504$)
BioHackathon Japan 2026

Class II / DRB bundle graph

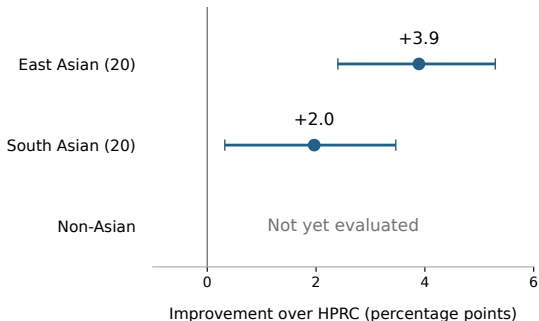


753 haplotypes; colour: bundle sharing (purple to yellow)

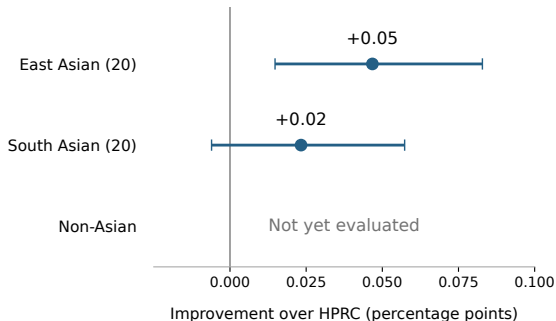
Variant recovery compared with HPRC

Larger gains for structural variants; small changes for SNVs

Structural variants



SNVs



Same reads and caller; expanded panel versus HPRC-only. Points: paired gains; bars: 95% intervals.

SV-bearing genotypes and shared SNV sites; horizontal scales differ. Test families excluded from panels; test assemblies remain in the graph.



T1K: independent evaluation

Correct diploid HLA genotypes / eligible truth

Truth cohort	Two fields	Four fields
Assembly, 64 donors	97.2% 495/509	50.5% 163/323
Experimental, 946 donors	93.9% 4,437/4,723	Unavailable

Assembly cohort: 16 donors each from East Asian, South Asian, European and African strata; eight HLA genes.

Experimental cohort: Gourraud, five HLA genes; two-field truth only. One failed donor remains in the denominator.

IPD 3.65. Failed and unresolved calls count as incorrect; eligible truth subsets differ by resolution.